

Java Programming Assignment Report

Name: Muthyala Sarayu

College: G Narayanamma Institute of Technology and Science

Branch: Electronics and Communication Engineering (ECE)

Task 1

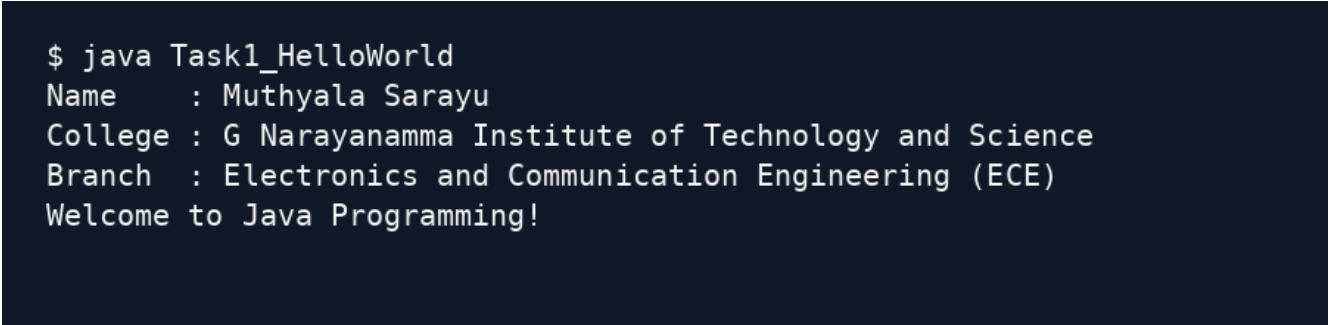
Brief Explanation

This program demonstrates the basic structure of a Java application and prints personal and college information using `System.out.println()`.

Code Snippet

```
public class Task1_HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Name      : Muthyala Sarayu");  
        System.out.println("College   : G Narayanamma Institute of Technology and Science");  
        System.out.println("Branch   : Electronics and Communication Engineering (ECE)");  
        System.out.println("Welcome to Java Programming!");  
    }  
}
```

Output Screenshot

A screenshot of a terminal window with a dark background. It shows the command to run a Java class and its output. The command is '\$ java Task1_HelloWorld'. The output consists of four lines: 'Name : Muthyala Sarayu', 'College : G Narayanamma Institute of Technology and Science', 'Branch : Electronics and Communication Engineering (ECE)', and 'Welcome to Java Programming!'.

```
$ java Task1_HelloWorld  
Name      : Muthyala Sarayu  
College   : G Narayanamma Institute of Technology and Science  
Branch   : Electronics and Communication Engineering (ECE)  
Welcome to Java Programming!
```

Task 2

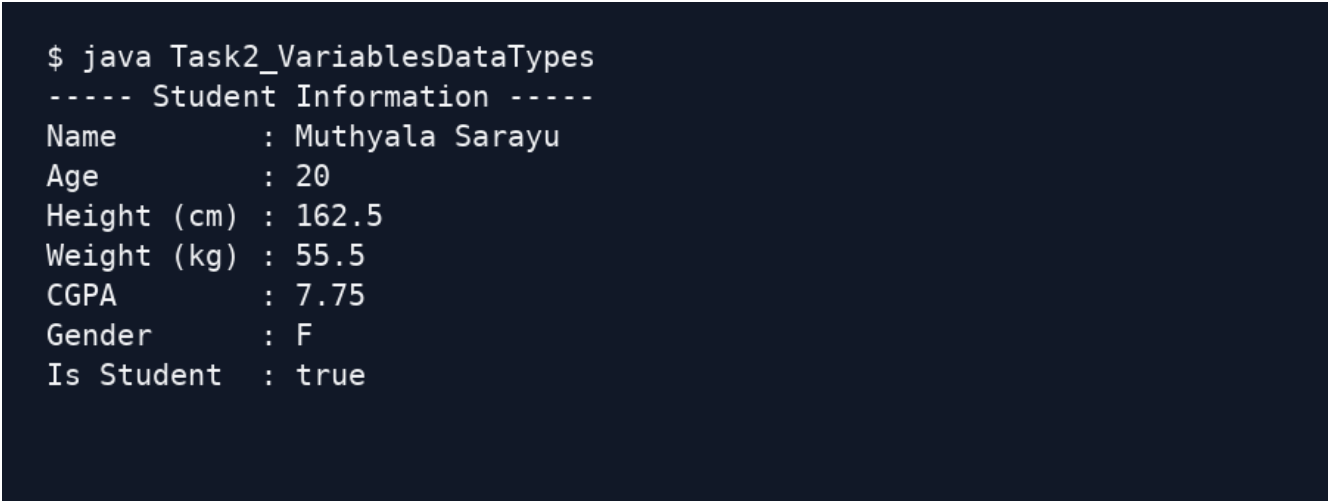
Brief Explanation

This program demonstrates common Java data types: String, int, float, double, char, and boolean. The values are displayed in a neat format.

Code Snippet

```
public class Task2_VariablesDataTypes {  
    public static void main(String[] args) {  
        String name = "Muthyala Sarayu";  
        int age = 20;  
        float height = 162.5f;  
        double weight = 55.5;  
        double cgpa = 7.75;  
        char gender = 'F';  
        boolean isStudent = true;  
  
        System.out.println("----- Student Information -----");  
        System.out.println("Name          : " + name);  
        System.out.println("Age           : " + age);  
        System.out.println("Height (cm)  : " + height);  
        System.out.println("Weight (kg)  : " + weight);  
        System.out.println("CGPA         : " + cgpa);  
        System.out.println("Gender       : " + gender);  
        System.out.println("Is Student   : " + isStudent);  
    }  
}
```

Output Screenshot

A screenshot of a terminal window with a dark background. It shows the execution of a Java program. The first line is the command to run the class. The subsequent lines show the output of the program, which is a formatted display of student information.

```
$ java Task2_VariablesDataTypes  
----- Student Information -----  
Name          : Muthyala Sarayu  
Age           : 20  
Height (cm)   : 162.5  
Weight (kg)   : 55.5  
CGPA          : 7.75  
Gender        : F  
Is Student    : true
```

Task 3

Brief Explanation

The Scanner class is used to accept text and numeric input from the keyboard. `nextLine()` reads strings, while `nextInt()` reads the age.

Code Snippet

```
import java.util.Scanner;

public class Task3_ScannerInput {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter Full Name: ");
        String fullName = scanner.nextLine();

        System.out.print("Enter Age: ");
        int age = scanner.nextInt();
        scanner.nextLine(); // Consume the leftover newline.

        System.out.print("Enter College Name: ");
        String collegeName = scanner.nextLine();

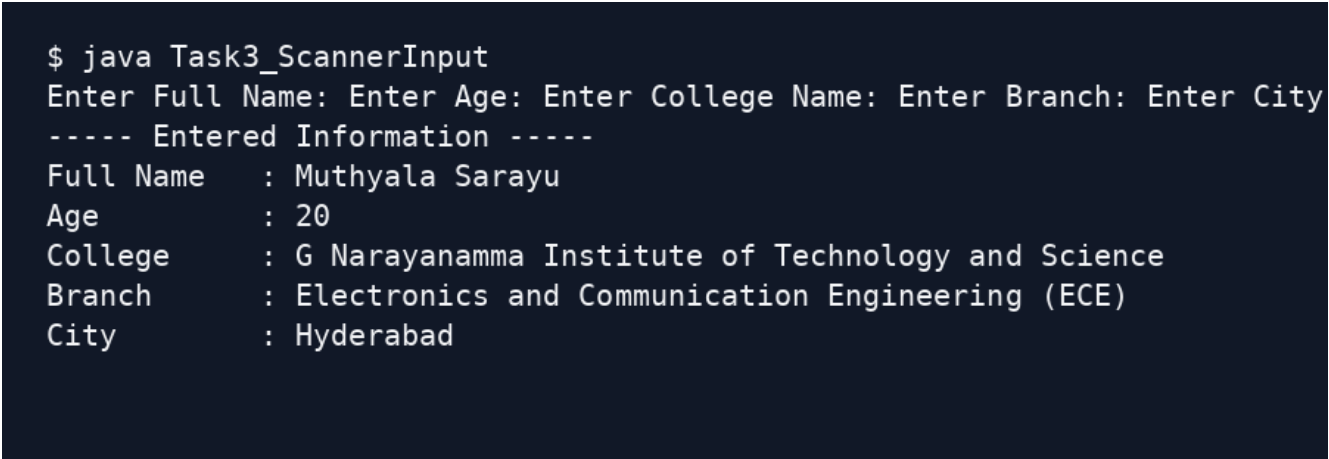
        System.out.print("Enter Branch: ");
        String branch = scanner.nextLine();

        System.out.print("Enter City: ");
        String city = scanner.nextLine();

        System.out.println("\n----- Entered Information -----");
        System.out.println("Full Name    : " + fullName);
        System.out.println("Age      : " + age);
        System.out.println("College   : " + collegeName);
        System.out.println("Branch    : " + branch);
        System.out.println("City      : " + city);

        scanner.close();
    }
}
```

Output Screenshot



```
$ java Task3_ScannerInput
Enter Full Name: Enter Age: Enter College Name: Enter Branch: Enter City
----- Entered Information -----
Full Name    : Muthyala Sarayu
Age          : 20
College      : G Narayanamma Institute of Technology and Science
Branch       : Electronics and Communication Engineering (ECE)
City         : Hyderabad
```

Task 4

Brief Explanation

The calculator accepts two numbers and demonstrates arithmetic operators: +, -, *, /, and %. A zero divisor is handled safely.

Code Snippet

```
import java.util.Scanner;

public class Task4_Calculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double firstNumber = scanner.nextDouble();

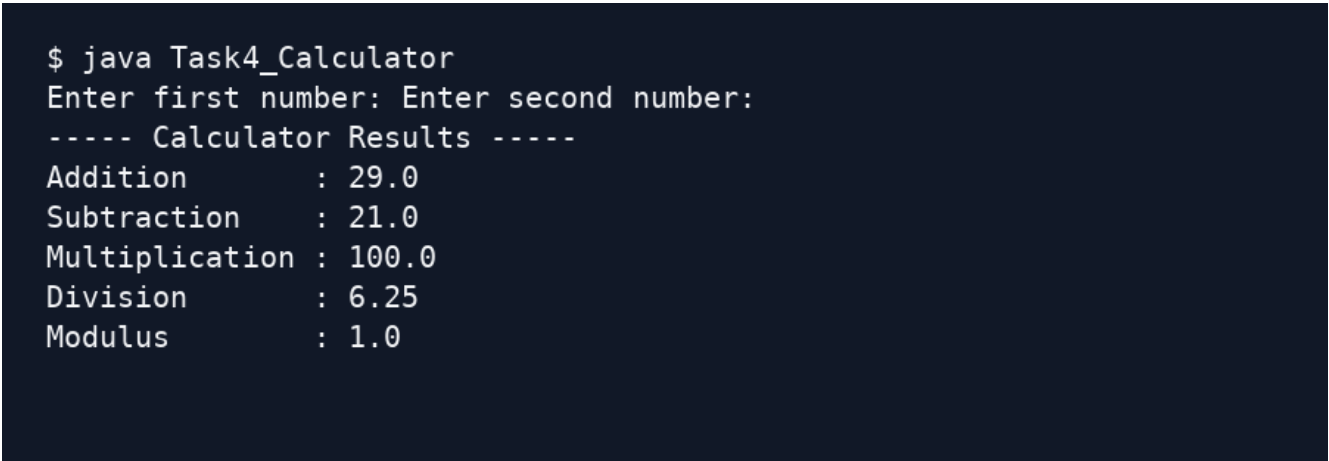
        System.out.print("Enter second number: ");
        double secondNumber = scanner.nextDouble();

        System.out.println("\n----- Calculator Results -----");
        System.out.println("Addition      : " + (firstNumber + secondNumber));
        System.out.println("Subtraction  : " + (firstNumber - secondNumber));
        System.out.println("Multiplication : " + (firstNumber * secondNumber));

        if (secondNumber != 0) {
            System.out.println("Division      : " + (firstNumber / secondNumber));
            System.out.println("Modulus       : " + (firstNumber % secondNumber));
        } else {
            System.out.println("Division      : Cannot divide by zero");
            System.out.println("Modulus       : Cannot calculate modulus by zero");
        }

        scanner.close();
    }
}
```

Output Screenshot



```
$ java Task4_Calculator
Enter first number: Enter second number:
----- Calculator Results -----
Addition      : 29.0
Subtraction   : 21.0
Multiplication : 100.0
Division      : 6.25
Modulus       : 1.0
```

Task 5

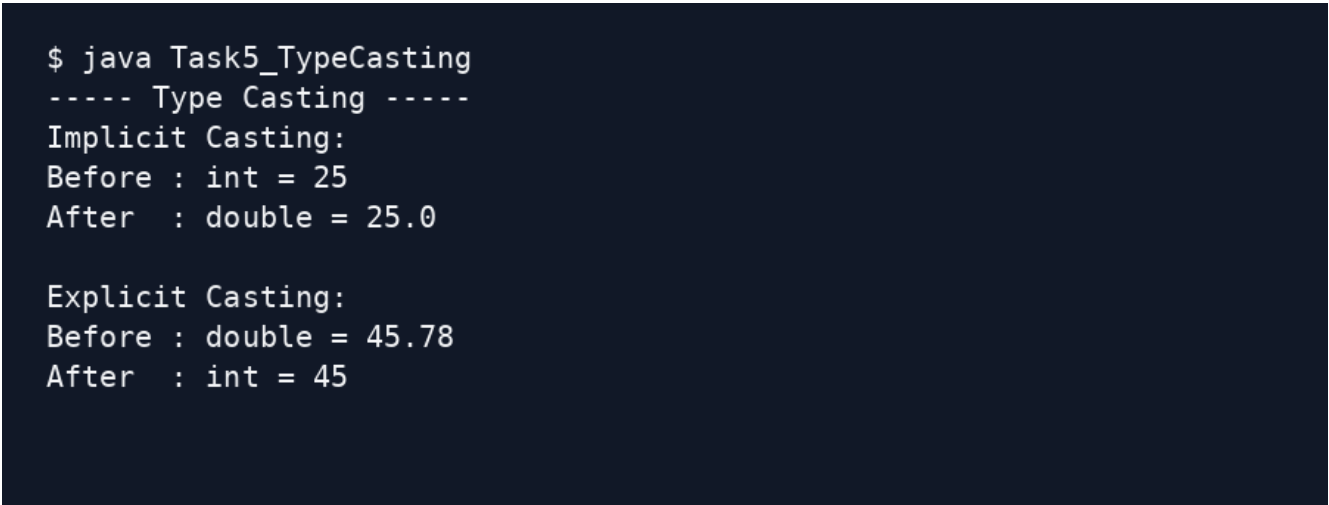
Brief Explanation

Implicit casting is automatic widening conversion from a smaller compatible type to a larger type, such as int to double. Explicit casting is manual narrowing conversion, such as double to int, using a cast operator. Explicit casting may lose the fractional part or other information.

Code Snippet

```
public class Task5_TypeCasting {  
    public static void main(String[] args) {  
        // Implicit casting: int is automatically converted to double.  
        int integerValue = 25;  
        double implicitValue = integerValue;  
  
        // Explicit casting: double is manually converted to int.  
        double decimalValue = 45.78;  
        int explicitValue = (int) decimalValue;  
  
        System.out.println("----- Type Casting -----");  
        System.out.println("Implicit Casting:");  
        System.out.println("Before : int = " + integerValue);  
        System.out.println("After  : double = " + implicitValue);  
  
        System.out.println("\nExplicit Casting:");  
        System.out.println("Before : double = " + decimalValue);  
        System.out.println("After  : int = " + explicitValue);  
    }  
}
```

Output Screenshot



```
$ java Task5_TypeCasting  
----- Type Casting -----  
Implicit Casting:  
Before : int = 25  
After  : double = 25.0  
  
Explicit Casting:  
Before : double = 45.78  
After  : int = 45
```

Short Explanation

Implicit casting happens automatically when converting a smaller compatible numeric type to a larger type. Explicit casting requires writing the target type in parentheses before the value. Explicit casting is useful for narrowing conversions but may lose information, such as decimal digits.

Task 6

Brief Explanation

Type casting is commonly used when converting data between numeric types, processing user input, performing calculations, and matching method parameter or return types. Widening conversions can happen automatically, while narrowing conversions usually require an explicit cast.

Code Snippet

```
import java.util.Scanner;

public class Task6_FundamentalsCasting {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter an integer value: ");
        int integerValue = scanner.nextInt();

        System.out.print("Enter a float value: ");
        float floatValue = scanner.nextFloat();

        System.out.print("Enter a double value: ");
        double doubleValue = scanner.nextDouble();

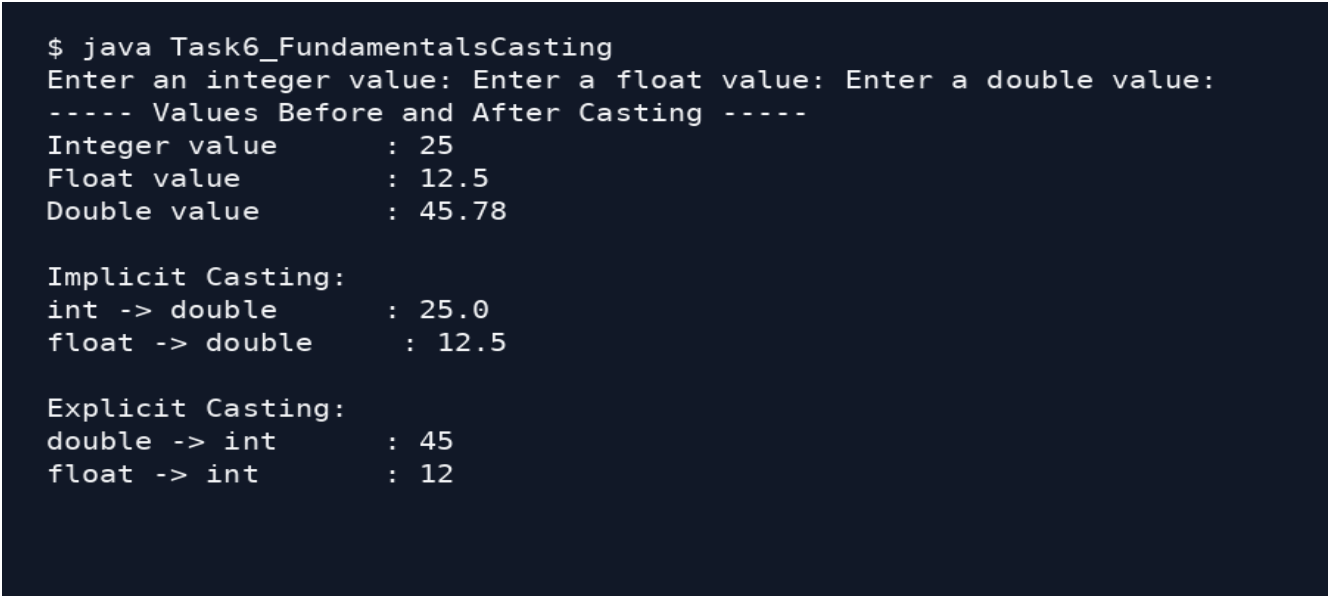
        // Implicit widening conversions happen automatically.
        double integerToDouble = integerValue;
        double floatToDouble = floatValue;

        // Explicit narrowing conversion requires a cast.
        int doubleToInteger = (int) doubleValue;
        int floatToInteger = (int) floatValue;

        System.out.println("\n----- Values Before and After Casting -----");
        System.out.println("Integer value      : " + integerValue);
        System.out.println("Float value       : " + floatValue);
        System.out.println("Double value      : " + doubleValue);
        System.out.println("\nImplicit Casting:");
        System.out.println("int -> double     : " + integerToDouble);
        System.out.println("float -> double    : " + floatToDouble);
        System.out.println("\nExplicit Casting:");
        System.out.println("double -> int      : " + doubleToInteger);
        System.out.println("float -> int       : " + floatToInteger);

        scanner.close();
    }
}
```

Output Screenshot



```
$ java Task6_FundamentalsCasting
Enter an integer value: Enter a float value: Enter a double value:
----- Values Before and After Casting -----
Integer value      : 25
Float value       : 12.5
Double value      : 45.78

Implicit Casting:
int -> double     : 25.0
float -> double    : 12.5

Explicit Casting:
double -> int      : 45
float -> int       : 12
```

Short Note

Type casting is commonly used when converting data between numeric types, processing user input, performing calculations, and matching method parameter or return types. Widening conversions can happen automatically, while narrowing conversions usually require an explicit cast.